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BUILDING USING MAVEN

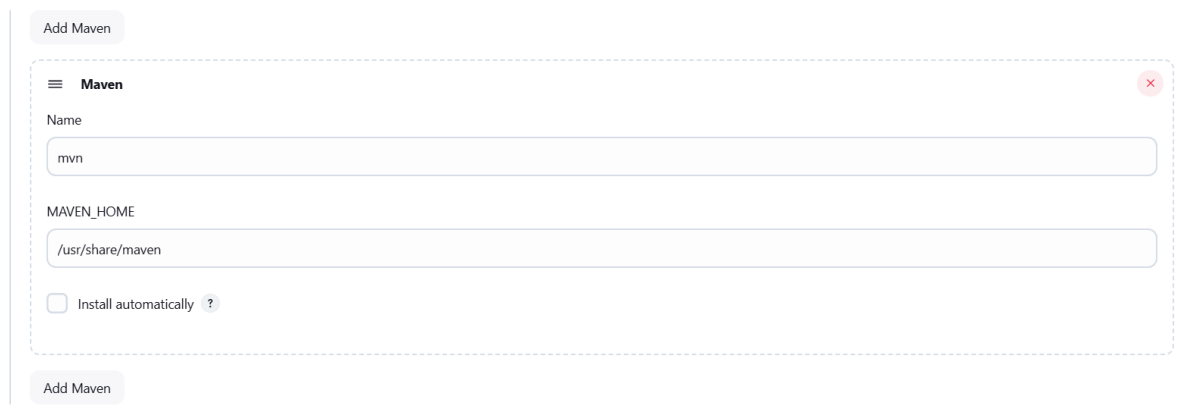
INTRODUCTION

Maven is a powerful build automation tool primarily used for Java projects. It simplifies project management by handling dependencies, builds, testing, and deployments in a structured and efficient manner.

Steps to build using maven

1. Configure maven in Jenkins

- Go to Jenkins Dashboard → Manage Jenkins → Tool Configuration
- Scroll to the Maven section and click Add Maven
- Provide a name and local path of the maven



2. Create a Maven Project in Jenkins

- Go to Jenkins Dashboard → New Item
- Select Freestyle Project
- Under Source Code Management, select Git and provide the repository URL

Source Code Management

☐ None
☒ Git ?

Repositories ?

Repository URL ? ✕

Credentials ?

▼

+ Add

Advanced ▼

Add Repository

Branches to build ?

Build Triggers

- ☐ Trigger builds remotely (e.g., from scripts) ?
☐ Build after other projects are built ?
☐ Build periodically ?
☒ GitHub hook trigger for GITScm polling ?
☐ Poll SCM ?

Build Steps

≡ **Invoke top-level Maven targets** ? ✕

Maven Version

▼

Goals

▼

Advanced ▼

Add build step ▼

3. Navigating Jenkins Workspace and Verifying Maven Build Artifacts

- Navigate to the Jenkins Workspace: `cd /var/lib/jenkins/workspace?Task_5`
- List Available Projects: `ls #` List all available projects in the Jenkins workspace
- Enter the Maven Project Directory: `cd Task_5`
- Enter the target Directory: `cd target`
- Verify the Generated Artifacts: `ls #` List the generated files (e.g., .jar or .war)

```
PS C:\Users\mathi> ubuntu
karunyamathivanan@LAPTOP-N670L8SV:~$ cd /var/lib/jenkins/workspace/Task_5
karunyamathivanan@LAPTOP-N670L8SV:/var/lib/jenkins/workspace/Task_5$ ls
LICENSE  README.md  deploymentservice.yml  docker  pom.xml  scripts  src  target
```

```
karunyamathivanan@LAPTOP-N670L8SV:/var/lib/jenkins/workspace/Task_5$ cd target
karunyamathivanan@LAPTOP-N670L8SV:/var/lib/jenkins/workspace/Task_5/target$ ls
classes      generated-test-sources  maven-status  shopping-cart-0.0.1-SNAPSHOT.jar.original
generated-sources  maven-archiver          shopping-cart-0.0.1-SNAPSHOT.jar  test-classes
```

4. Building, Pushing, and Deploying a Maven Project with Docker and Kubernetes

- Build Docker Image: `docker build -t test -f docker/Dockerfile .`
- Push Docker Image to Docker Hub : `docker push karunya08/Task_5`
- Create Kubernetes Deployment: `kubectl create deployment maven --image=test --port=80`
- Expose Deployment as a Service: `kubectl expose deployment maven --type=NodePort --port=80 --target-port=8070`
- Verify Docker Image: `docker images | grep karunya08/Task_5`
- Check Running Kubernetes Pods: `kubectl get pods`
- Get Service URL in Minikube: `minikube service maven`

```
Build an image from a Dockerfile
karunyamathivanan@LAPTOP-N670L8SV:/var/lib/jenkins/workspace/Task_5$ docker build -t test -f docker/Dockerfile .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon 49.2MB
Step 1/6 : FROM openjdk:8u151-jdk-alpine3.7
8u151-jdk-alpine3.7: Pulling from library/openjdk
ff3a5c916c92: Pull complete
a8906544047d: Pull complete
ae9db8d675e1: Pull complete
Digest: sha256:bd4030dd52cfb61a61e622fe74483e4e2089a2ef3d782bbf742a0d99093dal19
Status: Downloaded newer image for openjdk:8u151-jdk-alpine3.7
--> cc2179b8f042
Step 2/6 : EXPOSE 8070
--> Running in 821e533f398e
--> Removed intermediate container 821e533f398e
--> e1af71d95f09
Step 3/6 : ENV APP_HOME /usr/src/app
--> Running in 9b63d3bdb7a2
--> Removed intermediate container 9b63d3bdb7a2
--> 794d12c17f47
Step 4/6 : COPY target/shopping-cart-0.0.1-SNAPSHOT.jar $APP_HOME/app.jar
--> c71028b5753c
Step 5/6 : WORKDIR $APP_HOME
--> Running in 5a576978fd2d
--> Removed intermediate container 5a576978fd2d
--> 325f66171fca
Step 6/6 : ENTRYPOINT exec java -jar app.jar
--> Running in 764e8ec47af5
--> Removed intermediate container 764e8ec47af5
--> 616049e775de
Successfully built 616049e775de
Successfully tagged test:latest
karunyamathivanan@LAPTOP-N670L8SV:/var/lib/jenkins/workspace/Task_5$ docker tag test karunya08/task5
karunyamathivanan@LAPTOP-N670L8SV:/var/lib/jenkins/workspace/Task_5$ kubectl create deployment maven --image=karunya08/task5 --port 8070
deployment.apps/maven created
```

```
karunyamathivanan@LAPTOP-N670L8SV:/var/lib/jenkins/workspace/Task_5$ kubectl expose deployment maven --type=NodePort --port=8070 --target-port=8070
service/maven exposed
```

```
See 'snap info <snapname>' for additional versions.
karunyamathivanan@LAPTOP-N670L8SV:/var/lib/jenkins/workspace/Task_5$ kubectl get pods
NAME                                READY    STATUS    RESTARTS   AGE
dev-7bcb684f4c-xzhg5                1/1      Running   2 (45h ago) 2d19h
maven-595b7bfd48-2ch8m               1/1      Running   0           31s
my-deployment-d44754fb6-wvzmb        1/1      Running   1 (45h ago) 47h
test-7df658dfdd-8pgqf               1/1      Running   0           43h
```

```
karunya@karunya08:~$ minikube service maven
NAMESPACE   NAME   TARGET PORT   URL
default     maven   8070          http://192.168.49.2:32085
Starting tunnel for service maven.
NAMESPACE   NAME   TARGET PORT   URL
default     maven   8070          http://127.0.0.1:44053
Opening service default/maven in default browser...
http://127.0.0.1:44053
! Because you are using a Docker driver on linux, the terminal needs to be open to run it.
```

OUTPUT

